Flooding the filter: a game-theoretic model of AI, submission pressure, and quality collapse

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Abstract

Generative AI has made it dramatically cheaper to produce a manuscript, while the capacity to review one has grown far more slowly. Submission volumes have surged to record levels, a rising share of peer reviews contains AI-generated text, and verification infrastructure has not kept pace. We model this asymmetry using a game-theoretic framework in which AI adoption raises submission volume and lowers drafting cost faster than it raises review capacity, while also making it easier for weak manuscripts to pass screening. We find three results. First, extreme slot scarcity can stabilise honest submission norms even under degraded screening, because the publication lottery becomes too hard to win. Second, gradual AI adoption can trigger a discrete tipping point into a low-quality equilibrium; once tipped, the system is self-sustaining, so recovery requires a large coordinated intervention, not incremental reform. Third, expanding publication capacity without scaling verification accelerates the tipping. These results suggest that institutional reform should prioritise verification—structured audits, submission quotas, and decoupled certification—rather than simply expanding publication capacity.

1 Introduction

Generative AI is reshaping the production of scientific knowledge at a pace that outstrips the capacity of existing institutions to absorb it. By September 2024, more than one-fifth of computer-science preprints on arXiv showed measurable evidence of large-language-model modification (Liang et al., 2025), and at least 13.5% of all biomedical abstracts published that year contained AI-assisted text (Kobak et al., 2025). Monthly submissions to arXiv reached a record 24,226 in October 2024, roughly 50% higher than in late 2022 (arXiv, 2024). While production has surged, verification has not kept pace: the rate at which researchers accept reviewer invitations has declined steadily over two decades (Meyerson et al., 2025), and between 6.5% and 16.9% of conference peer reviews now contain substantial AI-generated text (Liang et al., 2024). More than half of active researchers surveyed in 2025 reported using AI tools to write reviews, often against explicit journal policy (Naddaf, 2025).

This asymmetry between production and verification is not unique to publishing. Historically, technologies that accelerate the creation of knowledge have outpaced the institutions designed to evaluate it, and sustained progress has depended on building what Mokyr (2002) calls institutional "bridges" between production and verification (Walker, 2026). AI intensifies this pattern: it automates precisely the tasks that are cheapest to verify—drafting, formatting, stylistic polish—while leaving the expensive tasks—checking data, replicating analyses, evaluating genuine novelty—largely untouched (Walker, 2026; Checco et al., 2021). The result is a growing gap between what can be produced and what can be checked. At the population level, AI-adopting scientists publish substantially more but follow narrower research trajectories, with distinct topics contracting and follow-on engagement between researchers declining (Hao et al., 2026)—a dynamic consistent with March's (1991) observation that systems tend to "refine exploitation more rapidly than exploration, becoming effective in the short run but self-destructive in the long run." The question is not whether individual researchers behave ethically, but whether the *system* as a whole can absorb an asymmetric shock to production and verification.

While the conventional response to AI in science focuses on disclosure requirements, detection tools, and editorial guidelines, these measures address individual conduct and leave untouched a structural problem:

even if every author is honest, reviewers can scrutinise only a finite number of manuscripts carefully per cycle. Peer review is a shared resource under increasing strain: the global system already consumed over 100 million hours of researcher time each year before AI entered the picture (Kovanis et al., 2016; Aczél et al., 2021), and screening quality declines even when reviewers act in good faith, simply because competing demands leave too little time per paper (Bianchi et al., 2018). When submission volume rises faster than review capacity, screening degrades—not because of misconduct, but because of congestion (Gross and Bergstrom, 2019). Congested screening lowers the effective quality threshold, and a lower threshold raises the expected payoff from submitting weak work. We focus on this feedback between volume and screening quality as the central mechanism.

The consequences of this feedback are not merely individual but collective. When the review filter is imprecise, submitting everything regardless of quality can become individually rational (Thurner and Hanel, 2011), and prestige-maximising incentives can reward strategic over honest behaviour across a multi-journal system (Zollman et al., 2024). The honest-submission norm is fragile: it collapses when the cost of mimicking quality falls even slightly (Tiokhin et al., 2021)—precisely the shift that AI-assisted writing produces. "Publish or perish" incentives already act as a selection force that spreads high-volume, low-quality strategies through academic communities (Smaldino and McElreath, 2016; Ioannidis, 2005), and career structures that treat paper counts as a proxy for scientific value reinforce this dynamic (Casadevall and Fang, 2014; Stephan, 2012). AI also polishes manuscripts without improving their reliability: AI-assisted reviews award systematically higher scores (Latona et al., 2024; Liang et al., 2024), and automated detection remains unreliable (Liang et al., 2023). These trends suggest that AI raises submission productivity faster than it raises verification capacity (Noy and Zhang, 2023; Stokel-Walker and Van Noorden, 2023).

Because the outcome depends not on any single author's choice but on how submission strategies spread through a population of researchers responding to shared incentives, we frame the problem using evolutionary dynamics. The adoption of AI tools, the decision to submit marginal work, and the erosion of reviewing norms are social phenomena: they propagate through imitation, competitive pressure, and institutional culture (Smaldino and McElreath, 2016). Zhong et al. (2026) show that when AI adoption spreads through a population, individually rational choices—letting AI substitute for human effort—can erode the collective capacity for exploration, and that group-level institutional structures are needed to counteract this drift. Evolutionary game theory captures precisely this tension between individual incentives and population-level outcomes (Weibull, 1995; Hofbauer and Sigmund, 1998), making it a natural framework for studying whether honest submission norms can survive the spread of AI-assisted flooding.

We distinguish two bottlenecks in the publishing system and show that AI affects them asymmetrically. The first is verification capacity: the number of papers that reviewers can evaluate carefully in a given period, denoted K_{rev}. The second is allocation capacity: the number of papers the journal can publish per period, denoted K. AI raises submission volume V by reducing the marginal cost of drafting and resubmitting papers, lowering the effort cost c of each submission. It may also raise the probability λ that a weak manuscript passes peer review, because AI-assisted writing produces polished text that is harder to distinguish from genuine contributions on surface features alone. While production scales rapidly with AI adoption, verification does not: deep expertise, data and code checking, and replication remain human-intensive tasks, and journal policies in many fields explicitly restrict AI use in reviews (Naddaf, 2025). Dissemination and certification can in principle be separated—preprint servers make manuscripts available immediately while editorial endorsement follows independently (Ginsparg, 2011)—but the certification function is what our model analyses, and it is certification that AI adoption threatens.

We model this asymmetry in an evolutionary game-theoretic framework, drawing on strategic models of scientific communication (Zollman et al., 2024; Thurner and Hanel, 2011). We consider a population of authors who choose between two strategies: submitting only genuine work (the cooperative strategy) or submitting everything including low-quality manuscripts (the defecting strategy). A journal applies peer review, subject to both a review capacity K_{rev} and a publication slot constraint K. We introduce an AI adoption parameter $a \in [0, 1]$. Higher a raises submission volume V by a factor $(1 + \gamma a)$, raises review capacity K_{rev} more slowly by $(1 + \delta a)$ with $\delta < \gamma$, lowers submission cost c, and raises the probability λ that weak papers pass review. By assumption, submission volume grows faster than review capacity—formalising the asymmetry observed empirically. The net effect of a on the profitability of submitting a low-quality paper, $\Pi_{bad}(x, a) = r \cdot \bar{\rho}(x, a) \cdot \pi_0(x, a) - c(a)$, is positive: AI makes flooding more attractive.

We find three results. First, extreme slot scarcity can stabilise honest submission norms even under

degraded screening, because the publication lottery becomes too hard to win. Scarcity, paradoxically, acts as a deterrent—but AI gradually erodes this protection by lowering the submission cost $c(a)$, so the deterrent requires progressively tighter slot constraints as adoption deepens. Second, gradual AI adoption can trigger a discrete tipping point into a low-quality equilibrium; once tipped, the system is self-sustaining. Recovery requires a large coordinated intervention, not incremental reform: the level of a at which quality collapses is strictly lower than the level at which quality recovers, so the system exhibits hysteresis. Third, expanding publication capacity (raising K) without scaling verification makes the situation worse, not better. More slots raise the expected payoff from submitting marginal work; under AI adoption, where $c(a)$ is already low and $\lambda(a)$ already elevated, even a modest improvement in the publication lottery is enough to push the system across the tipping threshold.

While existing literatures each address part of this problem, none unifies them. Models of strategic submission do not treat AI as a structural shock that shifts multiple inputs at once (Thurner and Hanel, 2011; Zollman et al., 2024; Tiokhin et al., 2021). The empirical AI-in-science literature documents faster production, slower review growth, and higher false-positive risk, but does not connect these to a model with clear predictions about the long-run state of the system (Noy and Zhang, 2023; Liang et al., 2025; Latona et al., 2024). Neither strand addresses the possibility that submission pressure can grow gradually until it crosses a threshold, after which the system shifts suddenly to a low-quality state and resists recovery. The institutional design literature asks how gatekeeping should work (Ellison, 2002; Muller-Itten, 2022; Cotton, 2013) but does not ask how its proposals must change when drafting cost approaches zero and the screening signal degrades continuously with adoption. We close these gaps by connecting all four channels—$c(a)$ falling, V rising, K_{rev} lagging, and $\lambda(a)$ rising—to the long-run state of the publishing system, characterising conditions under which quality collapses suddenly and is hard to restore, and deriving which institutional levers can prevent or reverse that shift.

These results suggest that the focus of institutional reform should be on verification—structured audits, submission quotas, and decoupled certification—rather than simply more publication slots. The model suggests that effective intervention must target two primary levers: raising genuine review capacity K_{rev} and reducing the probability λ that weak papers pass review. These are not incremental reforms; they constitute a different institutional design, one oriented towards verification rather than volume.

The paper proceeds as follows. Sections 2 and 3 present the baseline model and its AI extension. Section 4 analyses how AI adoption shifts equilibria, erodes stability, and amplifies the slot expansion paradox. Section 5 translates the analytical results into an institutional agenda. Section 6 derives testable predictions, Section 7 discusses limitations, and Section 8 concludes. We provide agent-based simulations, welfare analysis, extended predictions, and model extensions in the Supplementary Material.

2 Baseline Model

Players and strategies. We consider a population of N authors submitting to a single journal over repeated periods. Each manuscript has binary quality: good ($q = 1$) with probability α, and bad ($q = 0$) with probability $1 - \alpha$. Under replicator dynamics, intermediate submission strategies are evolutionarily unstable and are eliminated in the mean-field limit, reducing the effective strategy space to two options (Weibull, 1995; Hofbauer and Sigmund, 1998). Under **selective submission** (OG), an author submits only when $q = 1$. Under **universal submission** (AS), an author submits regardless of quality, hoping that imperfect screening allows bad papers through. We write $x \in [0, 1]$ for the share of authors playing OG; the remaining $1 - x$ play AS.

Two bottlenecks. The publication process has two distinct chokepoints, and each creates a different strategic force. The first is **review capacity** K_{rev}: the number of papers that can receive careful scrutiny per period. Submission volume is $V(x) = N[1 - (1 - \alpha)x]$, which rises as the AS share grows. When V exceeds K_{rev}, reviewer time must be spread across more submissions: each paper receives less careful attention, and the effective depth of scrutiny per manuscript falls (Bianchi et al., 2018; Kovanis et al., 2016). We define the **screening intensity** as:

$$\eta(x) = \min\left(1,\ \frac{K_{rev}}{V(x)}\right). \tag{1}$$

The second bottleneck is **publication capacity** K: the number of slots the journal can fill per period. Papers that pass review form a peer-accepted pool $M(x)$; at most K are published. The probability that a peer-accepted paper is ultimately published—the **rationing probability**—is:

$$\bar{\rho}(x) = \min\left(1,\ \frac{K}{M(x)}\right). \tag{2}$$

Screening errors and the spam payoff. Peer review is an imperfect filter (Bornmann et al., 2010; Cicchetti, 1991). A good paper is falsely rejected with probability ϵ; a bad paper wrongly passes with probability λ. Both error rates apply at full scrutiny. When $\eta < 1$, reviewer time per paper is diluted: each submission receives only a fraction η of the careful attention the filter requires, so the effective pass rates rise above their full-scrutiny values (Bianchi et al., 2018):

$$\pi_0(x) = 1 - \eta(x)(1-\lambda), \qquad \pi_1(x) = 1 - \eta(x)\,\epsilon. \tag{3}$$

As $\eta \to 0$, both rates converge to 1: the journal loses all discriminatory power because reviewers cannot give any paper the attention the filter requires. The peer-accepted pool is $M(x) = N\alpha\,\pi_1(x) + N(1-\alpha)(1-x)\,\pi_0(x)$. Because OG and AS authors submit good papers on identical terms, the strategic difference reduces to the net value of submitting a bad paper:

$$\Pi_{bad}(x) = r \cdot \bar{\rho}(x) \cdot \pi_0(x) - c, \tag{4}$$

where r is the publication reward and c the submission cost. We work in the region $\lambda r < c < r(1-\epsilon)$: without congestion, submitting a bad paper is unprofitable, but the incentive reactivates once screening degrades—a dynamic analogous to adverse selection in Akerlof's market for lemons (Akerlof, 1970).

Dynamics. The two strategies differ only in how they treat bad papers. An OG author submits only when $q = 1$ (probability α), earning expected fitness $f_{OG} = \alpha[r\bar{\rho}\,\pi_1 - c]$. An AS author submits regardless, earning $f_{AS} = \alpha[r\bar{\rho}\,\pi_1 - c] + (1-\alpha)[r\bar{\rho}\,\pi_0 - c]$. The fitness advantage of AS over OG is therefore $f_{AS} - f_{OG} = (1-\alpha)\,\Pi_{bad}(x)$. Substituting into the standard two-strategy replicator equation $\dot{x} = x(1-x)[f_{OG} - f_{AS}]$ gives:

$$\dot{x} = -x(1-x)(1-\alpha)\,\Pi_{bad}(x). \tag{5}$$

When $\Pi_{bad} > 0$, AS spreads and x falls; when $\Pi_{bad} < 0$, OG spreads and x rises. Two forces compete as the AS share grows. First, rising volume degrades screening (η falls, π_0 rises), making it more attractive to submit bad papers. Second, rising volume floods the publication queue ($\bar{\rho}$ falls), making the publication lottery harder to win. While the screening-collapse force pushes the system towards full corruption, the rationing force can counteract it: when slot scarcity is extreme, the lottery deters flooding even after screening has failed—a mechanism we call *defence by congestion*.

Four long-run outcomes. The signs of Π_{bad} at the two boundary states determine the long-run outcome. When Π_{bad} is negative at both boundaries, flooding is unprofitable everywhere and the system converges to **Global Honesty** (GH, $x = 1$). When Π_{bad} is positive at both boundaries, flooding pays everywhere and the system collapses to **Global Corruption** (GC, $x = 0$). When $\Pi_{bad}(1) < 0$ but $\Pi_{bad}(0) > 0$, both extremes are stable and an unstable tipping threshold x^* separates their basins of attraction: this is **Bistability** (BS), where history determines the outcome and recovery from corruption requires a large intervention. The reverse sign pattern ($\Pi_{bad}(1) > 0$, $\Pi_{bad}(0) < 0$) produces **Stable Coexistence** (SC) at an interior x^*, sustained by slot rationing. A key structural observation is that the two boundaries are controlled by different parameters: while the honest boundary ($x = 1$) is secured by keeping λ low, the corrupt boundary ($x = 0$) is held by keeping K tight. The two bottlenecks can therefore be targeted separately as policy instruments—a point we return to in Section 4. Figure 1 illustrates the bistable case. Π_{bad} is positive near $x = 0$ (corruption reinforces itself) and negative near $x = 1$ (honesty reinforces itself), with an unstable tipping threshold at $x^* \approx 0.80$ separating the two regions. Which outcome prevails depends entirely on where the system starts relative to this threshold.

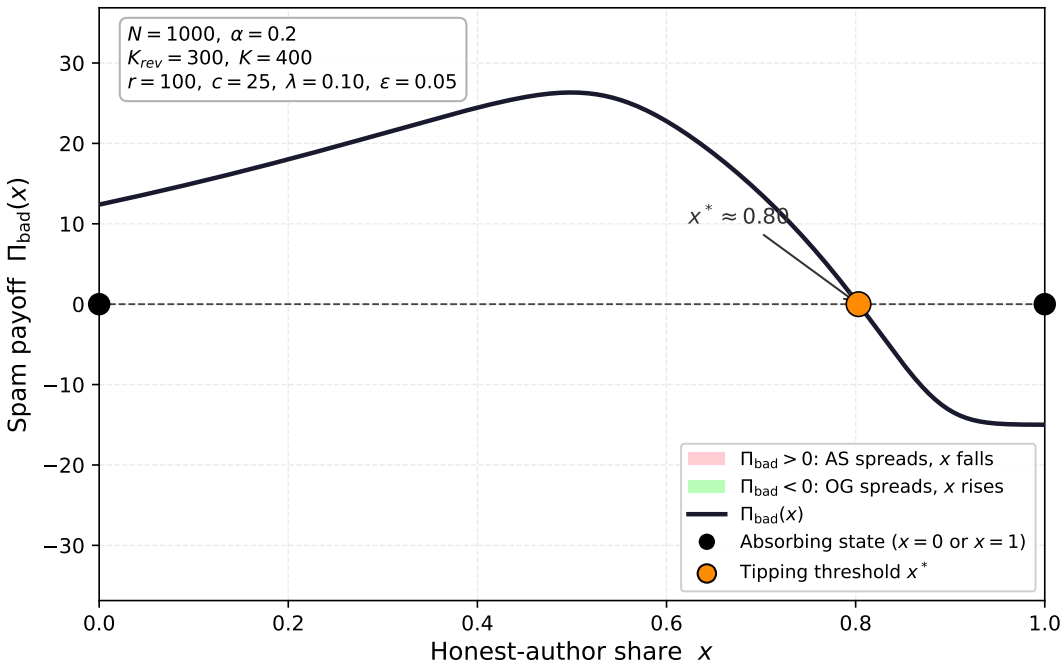

Figure 1: **Baseline phase portrait (bistability).** The curve plots $\Pi_{bad}(x)$, the net payoff from submitting a bad paper, against the honest-author share x. Parameters: $N = 1000$, $K_{rev} = 300$, $K = 400$, $r = 100$, $c = 25$, $\lambda = 0.1$. Red shading: $\Pi_{bad} > 0$, so AS spreads and x falls. Green shading: $\Pi_{bad} < 0$, so AS shrinks and x rises. Two long-run outcomes coexist—full honesty ($x = 1$) and full corruption ($x = 0$)—separated by an unstable tipping threshold at $x^* \approx 0.80$ (orange marker). Which outcome prevails depends entirely on the starting position relative to x^*.

3 The AI Extension

AI adoption as a parameter shift. Generative AI affects the publication system through multiple channels simultaneously. We capture this with a single scalar $a \in [0, 1]$, where $a = 0$ denotes the pre-AI baseline and $a = 1$ denotes full adoption across the author population. The parameter represents the field-level intensity of AI use—a composite of the fraction of authors employing AI tools and the depth of their integration into the writing and submission workflow. We hold a fixed and trace how the equilibrium structure from Section 2 changes as a deepens. As established in the Introduction, AI acts through four distinct channels, and the channels are not symmetric (see the mechanism diagram in the Supplementary Material).

Channel 1—Production: more submissions at every composition. The most direct effect of AI is to lower the marginal cost of drafting and preparing a manuscript, which raises total submission volume at any given composition of the author population. Randomised experiments find that LLM access cuts professional writing time by roughly 40% (Noy and Zhang, 2023; Stokel-Walker and Van Noorden, 2023). We model this as a proportional increase in total submission volume:

$$V(x, a) \;=\; N\big[1 - (1 - \alpha)x\big] \cdot (1 + \gamma a), \tag{6}$$

where $\gamma > 0$ governs how strongly AI adoption raises submission volume (the **production sensitivity**). System-level data confirm the scale of this shift: monthly arXiv submissions rose roughly 50% in two years, with over one-fifth of preprints showing measurable LLM modification (arXiv, 2024; Liang et al., 2025). V rises with a regardless of x: AS authors, whose constraint was drafting effort rather than ideas, now prepare and submit weak manuscripts they would not previously have bothered producing; OG authors face a lower marginal cost of bringing a manuscript to submission-ready quality, converting preliminary work into formal submissions faster. The underlying quality probability α of any individual

manuscript remains fixed; what changes is the number of submission-ready manuscripts each author produces per period.

Channel 2—Friction: lower cost of submitting bad papers. Beyond raising volume, AI directly lowers the cost of each submission attempt. We write:

$$c(a) \;=\; c_0(1 - \phi\, a), \qquad \phi \in (0, 1), \tag{7}$$

where c_0 is the baseline cost and ϕ governs the rate at which AI erodes submission friction. The cost reduction enters Π_{bad} directly: a lower c raises the net payoff from submitting a bad paper even if acceptance probabilities are unchanged. The effect is also asymmetric across strategies, though the asymmetry is not per-paper: c enters both Π_{bad} and Π_{good} symmetrically as a per-submission cost. The asymmetry arises from submission frequency. OG authors submit only when they have genuine work ready, so the total cost reduction they realise each period is small. AS authors submit everything—good papers and bad—so the same per-paper reduction compounds across a much larger submission stream, directly amplifying the flood incentive.

Channel 3—Vulnerability: weaker screening signal. AI-generated text is harder to distinguish on surface features (Stokel-Walker and Van Noorden, 2023), AI-assisted reviews award systematically higher scores (Latona et al., 2024; Liang et al., 2024), and automated detection remains unreliable—false-negative rates exceed 70% (Liang et al., 2023). Each mechanism raises the probability that a bad paper passes review. We model this as:

$$\lambda(a) \;=\; \lambda_0 + \mu\, a, \qquad \mu \geq 0, \tag{8}$$

where λ_0 is the baseline probability that a bad paper passes review and $\mu \geq 0$ governs how strongly AI adoption degrades screening quality. This specification captures the net effect of AI adoption on screening reliability without modelling each mechanism separately. Rising λ enters $\pi_0 = 1 - \eta(1 - \lambda)$ directly: even a carefully reviewed paper is more likely to pass.

Channel 4—Review assistance: capacity grows, but more slowly. AI can also assist reviewers—summarising papers, flagging inconsistencies, checking statistical reporting—so review capacity is not fixed. We model this as:

$$K_{rev}(a) \;=\; K_{rev}(0) \cdot (1 + \delta\, a), \tag{9}$$

where $\delta \geq 0$ governs how strongly AI adoption raises review capacity (the **verification sensitivity**). However, deep verification remains human-intensive: confidentiality rules restrict AI use in reviews (Naddaf, 2025; Checco et al., 2021), and reviewer capacity was already strained before the AI era (Meyerson et al., 2025). These constraints enforce what we call the **AI asymmetry assumption**:

$$\gamma > \delta \geq 0. \tag{10}$$

This is not an axiom but a testable empirical statement about relative scaling rates: submission volume grows faster with AI adoption than review capacity does.

Figure 2 summarises the four channels and their functional forms.

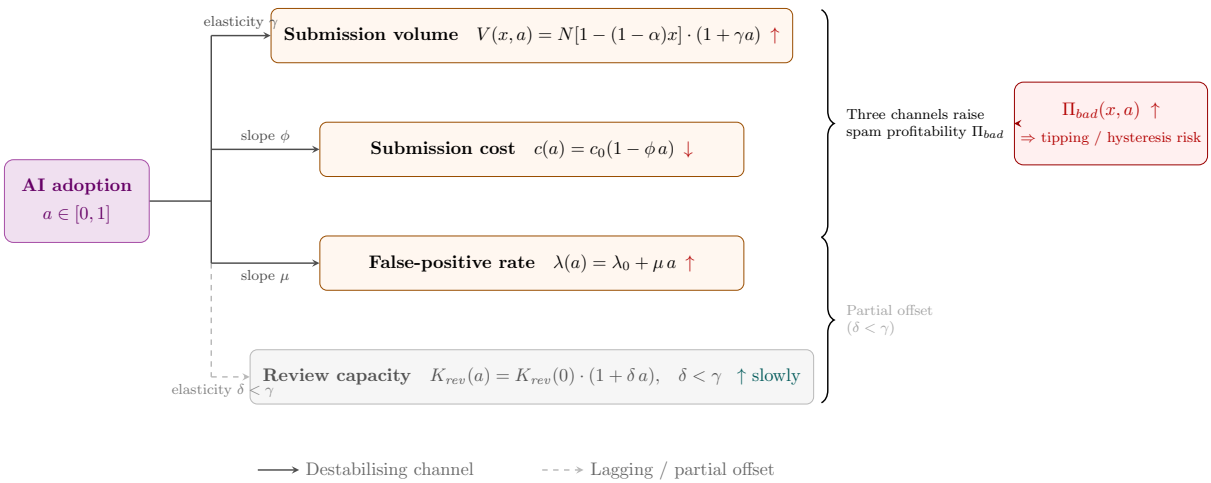

Figure 2: **AI adoption shifts four model parameters (Figure 3).** Solid arrows indicate channels that directly raise spam profitability Π_{bad}: rising submission volume (V, elasticity γ), falling submission cost (c, slope ϕ), and rising false-positive rate (λ, slope μ). The dashed arrow indicates the only offsetting channel—review capacity (K_{rev}, elasticity δ)—which grows more slowly than submission volume under the asymmetry assumption $\gamma > \delta$. All four channel equations are formal specifications of the informal channels described in the Introduction.

Putting the channels together: flooding becomes more profitable. Substituting the four channels into the baseline model yields the AI-extended spam payoff:

$$\Pi_{bad}(x,a) \;=\; r\cdot\bar{\rho}(x,a)\cdot\pi_0(x,a) \;-\; c(a), \tag{11}$$

where $\eta(x,a) = \min(1, K_{rev}(a)/V(x,a))$, $\pi_0(x,a) = 1-\eta(x,a)(1-\lambda(a))$, $\bar{\rho}(x,a) = \min(1, K/M(x,a))$, and

$$M(x,a) \;=\; (1+\gamma a)\big[N\alpha\,\pi_1(x,a) + N(1-\alpha)(1-x)\,\pi_0(x,a)\big]. \tag{12}$$

The factor $(1+\gamma a)$ reflects that AI inflates the total pool of papers reaching the publication queue in proportion to the same volume multiplier that governs $V(x,a)$: higher submission rates translate into a larger peer-accepted pool for any given pass rates π_0, π_1. This introduces a countervailing stabilising force. A larger M tightens slot rationing ($\bar{\rho} = K/M$ falls), reducing each spammer's probability of publication. The very flood of submissions that degrades screening also clogs the publication queue: $V\uparrow \Rightarrow\; M\uparrow \Rightarrow\; \bar{\rho}\downarrow \Rightarrow\; \Pi_{bad}\downarrow$. This congestion effect compresses the *revenue* term $r\bar{\rho}\pi_0$ rather than raising the submission cost $c(a)$, which falls with a. Whether the destabilising channels ($\pi_0\uparrow$, $c\downarrow$) or the stabilising channel ($\bar{\rho}\downarrow$) dominates depends on the relative magnitudes of K and M: when $K \gg M$, rationing is slack and the destabilising forces prevail; when $K \approx M$, the congestion defence is active. To isolate the core mechanism, consider the honest boundary $x = 1$, where only good papers are submitted and slot rationing is typically absent ($\bar{\rho}(1,a) \approx 1$ when publication slots exceed honest-author output):

$$\Pi_{bad}(1,a) \;\approx\; r\cdot\pi_0(1,a) \;-\; c(a). \tag{13}$$

At $x = 1$, only good papers are submitted, so submission volume equals $V(1,a) = N\alpha(1+\gamma a)$. For both calibration sets this remains strictly below $K_{rev}(a)$ for all $a \in [0,1]$, so $\eta(1,a) = 1$: review capacity is not the binding constraint at the honest boundary.[1] The bad-paper pass rate therefore reduces to $\pi_0(1,a) = \lambda(a)$, and the GH condition at $x = 1$ simplifies to $\lambda(a) < c(a)/r$. The critical mechanism at the honest boundary is not review congestion but the erosion of the screening signal relative to the cost of submission: $\lambda(a)$ rises while $c(a)/r$ falls simultaneously. Provided $\mu \geq 0$ and $\phi > 0$:

$$\frac{\partial \Pi_{bad}(1,a)}{\partial a} \;=\; r\;\underbrace{\mu}_{>0\text{ (rising }\lambda)}\;\underbrace{+\,c_0\phi}_{>0\text{ (falling }c)} \;>\; 0. \tag{14}$$

[1] For Set A: $K_{rev}(a) = 300(1+0.1a) > 200(1+0.5a) = V(1,a)$ requires $a < 10/7 \approx 1.43$, well above the maximum adoption level.

Setting $\Pi_{bad}(1, a^*) = 0$ gives a closed-form tipping point $a^* = (c_0 - r\lambda_0)/(r\mu + c_0\phi)$; for Set A, $a^* = (25-10)/(20+10) = 0.50$. Above a^*, $\Pi_{bad}(1, a) > 0$: the honest long-run outcome at $x = 1$ is no longer stable, and the system transitions from bistability into Global Corruption (or from GH into bistability if the corrupt boundary was previously deterred).

Creation versus verification: why AI is structurally asymmetric. The four-channel decomposition shows that AI exerts pressure from both the supply and quality dimensions simultaneously. While Channels 1 and 2 make it cheaper and faster to produce manuscripts, Channel 3 makes it harder to distinguish good submissions from bad ones at the review stage. Channel 4 is the only force working against the other three, but it is structurally constrained to grow more slowly. Writing is largely a syntactic task that language models perform well; verification is a semantic and domain-specific task requiring human expertise, data access, and contextual judgement (Checco et al., 2021). Institutions designed for the pre-AI era assumed rough parity between production and verification rates. That assumption no longer holds, and the asymmetry is not a temporary calibration issue—it reflects a fundamental difference in what AI can automate.[2]

Special cases: when stability can be preserved. Not every AI adoption trajectory leads to quality collapse. Equation (13) shows that the Global Honesty condition can persist even at high a if two circumstances hold jointly. First, if $\delta \approx \gamma$—review capacity scales nearly as fast as submission volume—then $\eta(1, a)$ does not fall and the congestion channel is neutralised. This could arise if structured AI-assisted verification pipelines (automated code checks, data-consistency audits) allow reviewers to handle more submissions without sacrificing depth. Second, if $\mu \approx 0$—AI tools do not degrade screening reliability, perhaps because structured reporting requirements close the gap that polished AI writing opens—then $\lambda(a)$ does not rise. When both conditions hold simultaneously, the cost reduction alone is typically insufficient to cross the tipping threshold, because $c(a)$ falls gradually while the honesty condition $c(a) > r\lambda(a)$ can remain satisfied. While the combination $\delta \approx \gamma$ and $\mu \approx 0$ keeps the system stable, the opposite case—review lagging severely ($\delta \ll \gamma$) and screening degrading ($\mu > 0$)—lets both channels reinforce each other and maximises the risk of irreversible tipping (see the scenario matrix in Appendix G of the Supplementary Material). The critical implication is that the institutional response determines which scenario the system occupies: while a is largely exogenous, δ and μ are policy choices.

Testable predictions. The AI extension generates four empirically distinguishable predictions. First, quality degradation should be *nonlinear* in AI adoption: the system holds near the honest equilibrium while a remains below the tipping threshold, then shifts abruptly rather than gradually. Smooth, continuous quality decline would be inconsistent with the bistable structure. Second, fields with higher AI adoption should show review-congestion signatures—longer turnaround times, rising desk-rejection rates, declining reviewer acceptance rates—as proxies for falling η (Meyerson et al., 2025). Third, acceptance rates should rise in fields with widespread AI-assisted reviewing, independently of any improvement in underlying research quality, as a signature of rising λ (Latona et al., 2024). Fourth, expanding publication slots should have *asymmetric* effects: while more slots increase access before tipping, after tipping they accelerate quality collapse by raising Π_{bad}. We examine the regime consequences of each lever in Section 4.

[2] We hold α (the probability a manuscript is genuinely good) fixed throughout the formal analysis. In principle, AI may also raise α via a quality-uplift channel: writing $\alpha(a) = \alpha_0(1 + \xi a)$ would add a partially offsetting force. Higher α increases the share of genuine work in the submission pool, which raises $M(x, a)$ and tightens slot rationing $\bar{\rho}$, thereby reducing Π_{bad}. However, the GH stability condition at $x = 1$ reduces to $\lambda(a) < c(a)/r$ when $\eta = 1$—a condition that does not involve α—so quality uplift does not shift the closed-form tipping threshold a^*. Evidence that AI raises genuine research quality is mixed and modest in magnitude (Noy and Zhang, 2023); we therefore treat α as fixed to focus on the four identified destabilising channels.

4 Equilibria, Tipping, and Hysteresis under AI Adoption

4.1 Equilibrium Conditions and the AI Shift

Whether the publishing system sustains honest norms or collapses into widespread flooding depends on the signs of Π_{bad} at two boundary states. At the *honest boundary* ($x = 1$), almost all authors are selective; the question is whether a single deviating author can profitably flood the system. At the *corrupt boundary* ($x = 0$), almost all authors submit everything; the question is whether the system can self-correct. These two boundaries are governed by different parameters. As we established in Section 2, the honest boundary is secured by keeping λ low, while the corrupt boundary is held by keeping K tight enough that the publication lottery deters flooding even under degraded screening.

We define publication quality as the share of genuinely good papers among those accepted:

$$Q(x) \;=\; \frac{N\alpha\,\pi_1(x)}{M(x)} \;=\; \frac{\alpha\,\pi_1(x)}{\alpha\,\pi_1(x) + (1-\alpha)(1-x)\,\pi_0(x)}. \tag{15}$$

Since slot rationing applies uniformly to the peer-accepted pool $M(x)$, it does not alter the ratio of good to bad papers; quality is therefore determined entirely by the composition x at the long-run outcome. The four possible long-run outcomes—GH, GC, BS, SC—are defined by the signs of Π_{bad} at the two boundary states, as described in the Baseline Model section.

Under AI adoption $a > 0$, four simultaneous shifts push $\Pi_{bad}(x, a)$ upward at every interior x. Rising submission volume (sensitivity γ) and lagging review capacity (sensitivity $\delta < \gamma$) lower the screening intensity $\eta(x, a)$, raising the bad-paper pass rate $\pi_0(x, a)$. Falling submission cost $c(a)$ directly raises the net payoff from each submission attempt. Rising $\lambda(a)$ further degrades the review filter even for papers that receive full scrutiny. The combined derivative (Equation 14) is strictly positive under the asymmetry $\gamma > \delta$: AI makes flooding more attractive at every composition of the author population. We now trace the structural consequences of this shift for regime boundaries, stability windows, and recovery conditions.

We use two representative parameter sets throughout (full calibration in Appendix G of the Supplementary Material). Set A ($K_{rev} = 300$, $K = 400$, $r = 100$, $c = 25$, $\lambda = 0.1$) represents a typical journal prone to bistability, where the review bottleneck is the binding constraint. Set B ($K_{rev} = 1200$, $K = 180$, $r = 150$, $c = 30$, $\lambda = 0.4$) represents a top-tier journal where extreme slot scarcity is the defining constraint. AI sensitivity parameters are fixed at $\gamma = 0.5$, $\delta = 0.1$, $\phi = 0.4$, $\mu = 0.2$ throughout.

4.2 How AI Shifts Regime Boundaries

Figure 3 presents the core comparative-statics result. The left panel shows the baseline ($a = 0$) regime map across the (K_{rev}, K) parameter space. The right panel shows the same map under high AI adoption ($a = 0.8$). The comparison is unambiguous: the Global Honesty (green) region shrinks substantially, the Bistability (yellow) and Global Corruption (red) zones expand and shift left, and the Stable Coexistence (blue) region contracts. A journal safely in the GH zone at baseline may find itself in BS or GC at $a = 0.8$ without any change in its own institutional parameters.

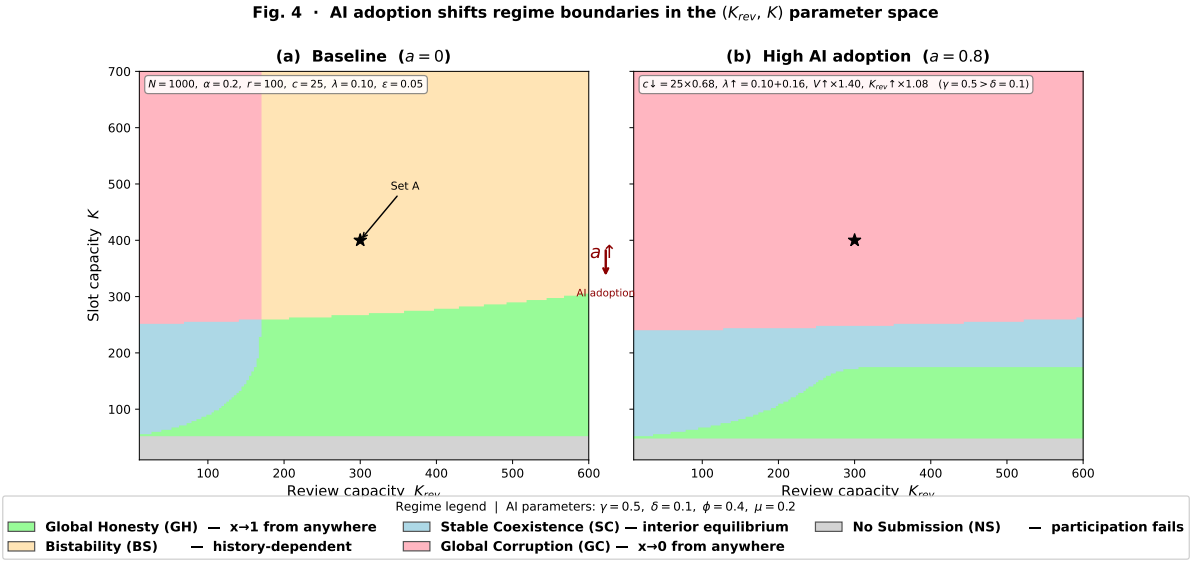

Figure 3: **AI adoption shifts regime boundaries in the (K_{rev}, K) parameter space (Fig. 4).** Left: baseline ($a = 0$). Right: high AI adoption ($a = 0.8$, $\gamma = 0.5$, $\delta = 0.1$, $\phi = 0.4$, $\mu = 0.2$). The Global Honesty (green) region shrinks and the Bistability and Corruption zones expand. The Set A calibration point (star) moves from a safe location at baseline towards the boundary of GH and BS under high AI adoption. The asymmetry $\gamma > \delta$ is the key driver: review capacity grows too slowly to absorb the AI-driven surge in submission volume.

The magnitude of this shift is quantifiable. At $a = 0.8$ with $\gamma = 0.5$ and $\delta = 0.1$, submission volume has risen by 40% while review capacity has risen by only 8%. A journal with $K_{rev} = 300$ now has the effective screening power of a journal with $K_{rev} \approx 231$ at baseline—a loss of nearly one-quarter of its screening capacity with no change in institutional design. Restoring the same screening intensity would require raising K_{rev} to approximately 389, a 30% increase in review resources.

The slot expansion paradox is amplified by AI. Expanding publication slots (K) seems like an obvious response to submission pressure, but we show that this response backfires when verification does not scale proportionally. Increasing K raises the rationing probability $\bar{\rho}(x, a)$, which raises the expected payoff from submitting a bad paper. AI amplifies this effect: when $c(a)$ is already low and $\lambda(a)$ already elevated, even a modest improvement in the publication lottery is enough to push $\Pi_{bad}(1, a)$ from negative to positive, tipping the system from GH into Bistability. More slots increase the prize in a game whose entry barrier has already been eroded by AI. This is visible in Figure 3: moving upward (increasing K) in the right panel is more dangerous than the same move in the left panel. The slot expansion paradox identified in the baseline model is not merely preserved under AI adoption—it is exacerbated.

Defence by congestion and its limits. Extreme slot scarcity can stabilise an otherwise fragile system. When K is small relative to the peer-accepted pool M, the rationing probability $\bar{\rho}$ is so low that the expected payoff from submitting a bad paper falls below the submission cost even under degraded screening—the journal's publication lottery is simply too hard to win. This defence operates through the condition $\bar{\rho}(0, a) \cdot \pi_0(0, a) < c(a)/r$, and all three terms move against it as a rises. Rising $\pi_0(0, a)$ from degraded screening and rising λ increase the left side; falling $c(a)$ reduces the right side. The net effect is that a journal relying on slot scarcity needs progressively tighter K to maintain the same deterrent as a increases. At some point, K must be so low that the participation constraint for honest authors—$\Pi_{good}(1, a) > 0$—is violated, pushing the system into the No Submission zone. While defence by congestion is a real stabilising force, it is not robust to sustained increases in a.

4.3 Equilibrium Trajectories and Stability Window Erosion

Figure 4 traces the long-run honest-author share $x^*(a)$ as AI adoption rises from 0 to 1 under three parameter scenarios.

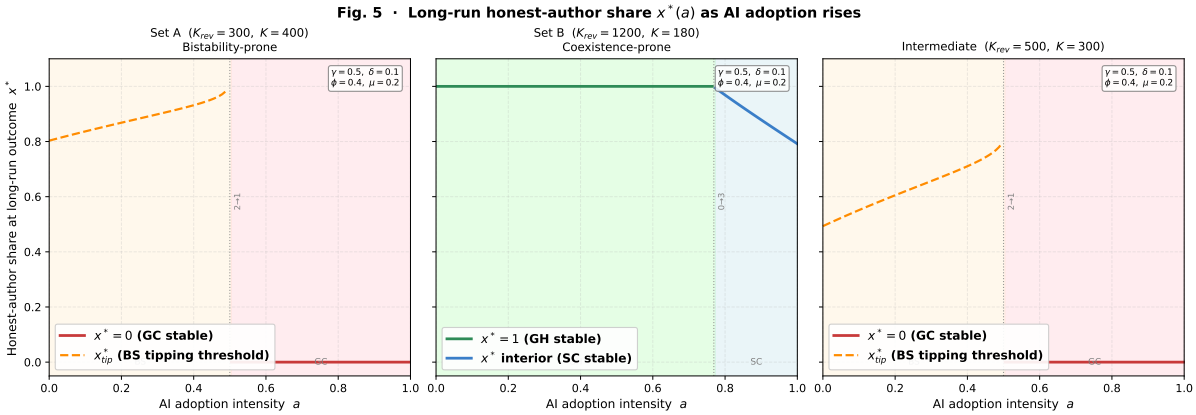

Figure 4: **Long-run honest-author share $x^*(a)$ as AI adoption rises (Fig. 5).** Left: Set A (bistability-prone, $N = 1000$). The system is already in BS at $a = 0$: the corrupt equilibrium is self-sustaining ($\Pi_{bad}(0,0) \approx +12.4 > 0$) while the honest equilibrium is stable ($\Pi_{bad}(1,0) = -15 < 0$), giving a tipping threshold $x^*_{tip}(0) \approx 0.80$ (orange dashed). As a rises, x^*_{tip} climbs monotonically towards $x = 1$ until the honest equilibrium collapses at $a \approx 0.50$ (BS→GC, vertical dashed line). The full GH→BS→GC transition under lower submission pressure ($N = 300$) is illustrated in Figure 6 (a). Centre: Set B (coexistence-prone). The interior stable long-run outcome x^* falls as a rises, tracking the declining profitability of honest submission relative to spamming. Right: Intermediate parameters. Shows the full GH→BS→GC transition as a increases. Vertical dashed lines mark regime transitions; shaded backgrounds indicate the active regime.

The left panel (Set A, $N = 1000$) shows a system that is already in Bistability at $a = 0$. With $N = 1000$ authors, the full submission load ($V(0,0) = 1000$) overwhelms the review capacity ($K_{rev} = 300$), yielding $\Pi_{bad}(0,0) \approx +12.4 > 0$: the corrupt equilibrium is self-sustaining from the outset. The honest equilibrium remains stable ($\Pi_{bad}(1,0) = -15 < 0$), so the system starts in BS with a tipping threshold at $x^*_{tip}(0) \approx 0.80$. As a rises, x^*_{tip} climbs monotonically towards $x = 1$: the honest norm becomes progressively more fragile even as published quality appears stable. At $a \approx 0.50$, $\Pi_{bad}(1,a) = 0$ and the honest equilibrium loses stability entirely; the system transitions from BS into Global Corruption. The practical implication is striking: the system can appear healthy—with the population well above x^*_{tip}—while the safety margin silently narrows. A single coordinated wave of low-quality submissions can push the population below x^*_{tip} and initiate a collapse that the system cannot reverse on its own. The full GH→BS→GC sequence, in which a community starts from Global Honesty before any AI adoption, is illustrated for $N = 300$ in Figure 6 (a).

The right panel (Intermediate parameters) illustrates the full transition from GH through BS to GC as a increases. In the GC zone, the long-run outcome is $x^* = 0$: spamming dominates from every starting point. The path from GH to GC passes through a BS corridor, during which the long-run outcome is history-dependent and recovery is still possible. The width of this corridor—the range of a values during which recovery remains feasible—depends on how quickly $\lambda(a)$ rises and how much $c(a)$ falls.

Figure 5 maps this dynamic in the (N, a) plane, treating the author population size N as the external pressure variable.

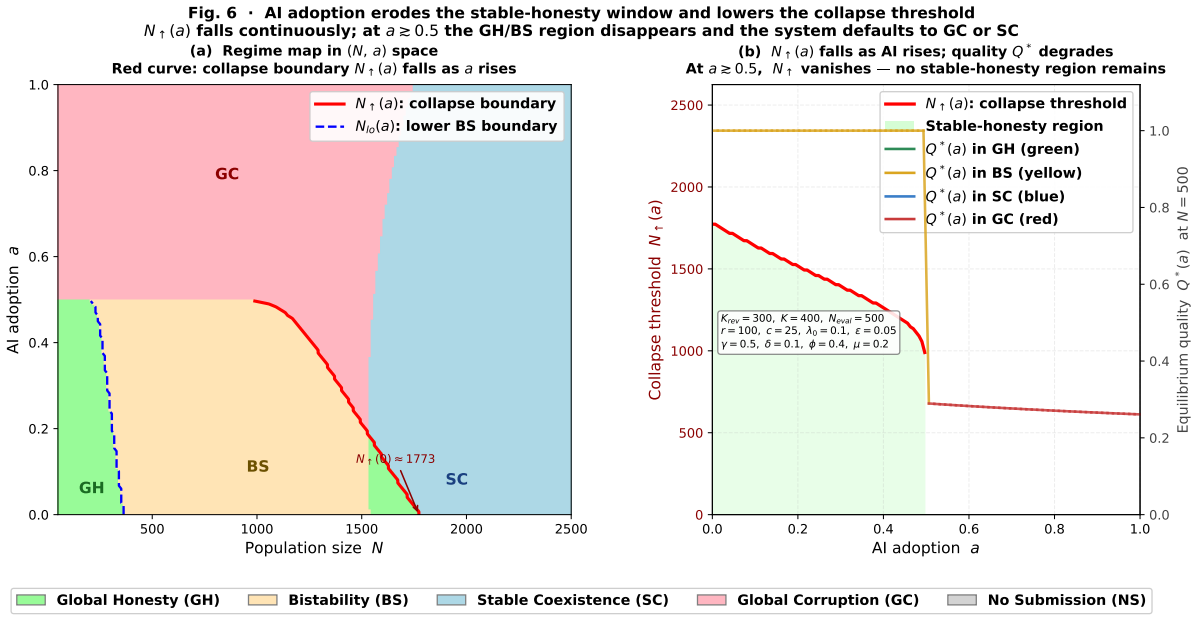

Figure 5: **AI adoption erodes the stable-honesty window and lowers the collapse threshold (Fig. 6).** Left: regime map in (N, a) space with Set A parameters. The red curve $N_\uparrow(a)$ marks the collapse boundary: to its right, the forward (honest-start) path collapses. The blue dashed curve marks the lower boundary of the Bistability zone. As a rises, $N_\uparrow(a)$ falls monotonically; at $a \gtrsim 0.5$ (for the chosen AI parameters) the GH and BS zones disappear entirely. Right: $N_\uparrow(a)$ (red, left axis) falls continuously; equilibrium quality $Q^*(a)$ at moderate pressure $N = 500$ (right axis, coloured by regime) also declines as a rises, tracking the regime transitions from BS (yellow) to GC (red); $N = 500$ lies in the bistable zone at $a = 0$ and crosses into Global Corruption near $a \approx 0.50$.

The left panel of Figure 5 shows the collapse threshold $N_\uparrow(a)$ falling monotonically as a rises. At $a = 0$, a population of $N \approx 1800$ is needed to push Set A out of the stable-honesty zone; at $a = 0.5$, this threshold has fallen to roughly half that. The right panel shows the corresponding quality decline: at a moderate load ($N = 500$), quality $Q^*(a)$ is near 1 at low a but falls continuously, tracking the regime transitions from GH through BS to SC. Crucially, at $a \gtrsim 0.5$, the bistable zone disappears entirely. There is no longer any starting position from which the honest norm is self-sustaining. Recovery from this state requires not merely a reduction in pressure N but a qualitative change in institutional parameters (K_{rev} raised, λ reduced, c restored). AI adoption, in this sense, transforms the challenge from maintenance to repair.

Figure 6 makes the hysteresis structure explicit as a classic fold bifurcation diagram, plotting the long-run honest-author share x^* against AI adoption a under two field-pressure levels.

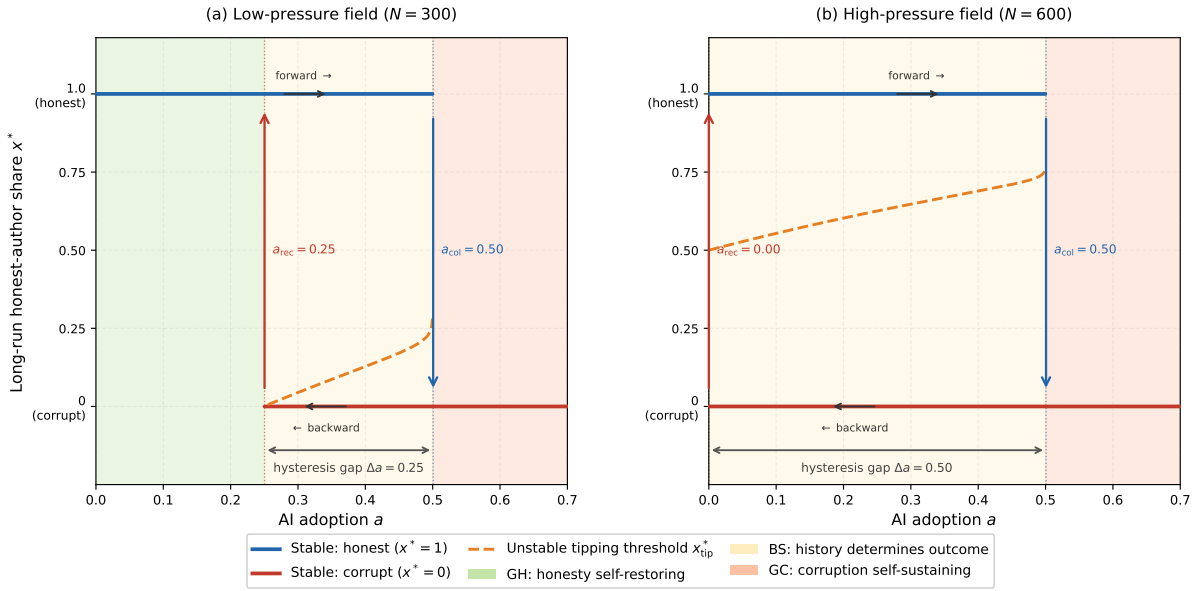

Figure 6: **Hysteresis loop: collapse and recovery thresholds in a-space (Fig. 6b).** Each panel plots the long-run honest-author share x^* against AI adoption a. **(a)** Low-pressure field ($N = 300$). The blue upper branch ($x^* = 1$) is stable while $a < a_{\text{col}} = 0.50$; the red lower branch ($x^* = 0$) is stable while $a > a_{\text{rec}} \approx 0.25$. The dashed orange curve is the unstable tipping threshold x^*_{tip} that separates the two basins in the Bistability corridor. A forward path (increasing a) follows the honest branch and collapses at a_{col}; a backward path (decreasing a) follows the corrupt branch and recovers only at a_{rec}, giving a hysteresis gap $\Delta a \approx 0.25$. **(b)** High-pressure field ($N = 600$). The system starts in Bistability at $a = 0$: the corrupt equilibrium is self-sustaining before any AI adoption. The honest branch still collapses at $a_{\text{col}} = 0.50$, but the corrupt branch is stable for all $a \in [0, 1]$: no recovery threshold exists in the admissible range.

In panel (a), the system occupies Global Honesty at baseline ($a = 0$), indicated by the green background. As a rises, the corrupt equilibrium first becomes self-sustaining at $a_{\text{rec}} \approx 0.25$, opening the Bistability corridor (yellow background); the honest equilibrium then loses stability at $a_{\text{col}} = 0.50$, pushing the system into Global Corruption (red background). While the forward path traces the upper branch to a_{col} and then drops discontinuously to $x^* = 0$, the backward path recovers only when a falls to a_{rec}—a full 25 percentage points below the collapse point. The hysteresis gap $\Delta a = a_{\text{col}} - a_{\text{rec}} \approx 0.25$ measures the depth of the trap: a field that has tipped must substantially reduce AI-linked submission pressure, not merely stop increasing it. The unstable tipping threshold (dashed) connects the two fold points; any perturbation large enough to push the system below x^*_{tip} triggers irreversible collapse so long as a remains above a_{rec}.

Panel (b) shows a qualitatively different situation. At $N = 600$, the corrupt equilibrium is already self-sustaining at $a = 0$: the system starts in Bistability rather than Global Honesty. The collapse threshold remains $a_{\text{col}} = 0.50$ (this value is independent of N when $\eta = 1$ at $x = 1$, since it is determined purely by $c(a) = r\lambda(a)$). However, there is no recovery threshold in the range $a \in [0, 1]$: the lower branch persists all the way back to $a = 0$, so reducing AI adoption alone cannot restore Global Honesty. Escaping the corrupt basin requires an institutional intervention that raises K_{rev}, reduces λ, or both, until $\Pi_{bad}(0, a)$ turns negative—a shift in the parameter configuration, not merely a reversal of a. The contrast between panels (a) and (b) captures the core asymmetry of the recovery problem: while low-pressure fields retain a restoration option, high-pressure fields require coordinated institutional action regardless of what happens to AI adoption.

4.4 The Screening Signal as Critical Destabiliser

Among the four AI channels, rising $\lambda(a)$ is structurally the most dangerous. The GH condition at $x = 1$ requires $\Pi_{bad}(1, a) < 0$; when screening is fully effective ($\eta = 1$), this simplifies to $\lambda < c(a)/r$. Under

baseline parameters ($c = 25$, $r = 100$), this requires $\lambda < 0.25$. A slope of $\mu = 0.2$ over full adoption ($a = 1$) pushes λ from 0.10 to 0.30, crossing this threshold and making GH structurally impossible regardless of review capacity. No investment in K_{rev} can restore GH once $\lambda \geq c/r$, because even a perfectly reviewed bad paper passes at rate λ and yields a positive expected payoff. This result is consistent with signalling theory: the reliability of the screening signal—not merely the volume of screening—sustains the separating equilibrium (Tiokhin et al., 2021; Ellison, 2002).

The screening-signal channel also explains why AI-assisted reviewing is particularly concerning. AI-assisted reviews award systematically higher scores, raising acceptance rates (Latona et al., 2024; Liang et al., 2024). If widespread AI use in reviewing raises the average λ, the effect is direct and structural: the GH boundary shifts inward regardless of what authors do. Interventions that target λ directly—structured reporting checklists, mandatory data and code deposition, statistical audit requirements—therefore address the most critical vulnerability in the system.

4.5 Policy Levers and Summary Propositions

Each parameter corresponds to a concrete institutional intervention. We summarise the mapping in four propositions; formal derivations are in Appendix A of the Supplementary Material.

Proposition 1 (Monotone destabilisation). *Under the asymmetry $\gamma > \delta \geq 0$: (i)* ***Honest boundary.*** *$\Pi_{bad}(1, a)$ is strictly increasing in a (analytically, for all parameter values satisfying the model assumptions). The Global Honesty region in (K_{rev}, K) space shrinks strictly as a rises, and there exists a finite closed-form threshold $a^* = (c_0 - r\lambda_0)/(r\mu + c_0\phi) \in (0, \infty)$ above which no (K_{rev}, K) combination sustains GH without additional institutional intervention. (ii)* ***Interior compositions.*** *For $x \in (0, 1)$, $\partial\Pi_{bad}/\partial a > 0$ holds analytically when review is uncongested ($\eta = 1$, Case I) or rationing is inactive ($\bar{\rho} = 1$, Case II). When both constraints bind (Case III), the result holds under a sufficient condition on the screening signal (stated in Appendix A of the Supplementary Material) and is confirmed numerically throughout $(x, a) \in (0, 1) \times [0, 1]$ for both calibrations in Appendix G of the Supplementary Material. All three cases jointly imply that the GH region shrinks as a rises.*

Proposition 2 (Slot expansion paradox). *Holding K_{rev} and a fixed, increasing K weakly raises $\Pi_{bad}(x, a)$ at every x. The marginal destabilising effect of an additional slot is strictly larger at higher a, because lower $c(a)$ and higher $\lambda(a)$ amplify the gain from winning the publication lottery. The paradox is therefore amplified, not attenuated, by AI adoption.*

Proposition 3 (Erosion of congestion defence). *The corrupt-boundary stability condition $\Pi_{bad}(0, a) < 0$ holds if and only if $K < K^*(a) \equiv c(a)M(0, a)/[r\pi_0(0, a)]$. When the cost-reduction slope satisfies $\phi \geq \gamma$, $K^*(a)$ is strictly decreasing in a for all $a \in [0, 1]$ analytically. When $\gamma > \phi$ (as in the calibrations of Appendix G of the Supplementary Material), $K^*(a)$ is strictly decreasing for $a \geq (\gamma - \phi)/(2\phi\gamma)$ analytically and is confirmed numerically to be strictly decreasing throughout $[0, 1]$ under both Sets A and B. Consequently, a journal that relied on passive slot scarcity for protection at baseline may need to actively reduce K to maintain the same deterrent as a rises.*

Proposition 4 (Policy lever ranking). *Among the primary levers, reducing λ (verification quality) and raising K_{rev} (review capacity) are unconditionally beneficial: they expand the GH region without side effects. Raising c (submission friction) restores the honesty condition but may deter some honest authors. Reducing K (tightening slots) provides passive deterrence but risks the No Submission zone. Expanding K alone is the only intervention that can make the system strictly worse.*

Early warning signals. The model identifies observable signatures that precede a tipping-point collapse. Rising submission-to-review ratios (V/K_{rev}) signal declining screening intensity η. Lengthening turnaround times and increasing desk-rejection rates indicate reviewer overload and the onset of rubber-stamping. Acceptance rates rising faster than improvements in underlying research quality signal a rising effective λ. In fields where AI adoption is high, these signals should appear before any observable quality decline, because the system can remain near $x \approx 1$ while the tipping threshold x^*_{tip} rises towards the

current honest-author share. To make this concrete: under Set A parameters, $x^* \approx 0.80$ at baseline, leaving an apparent safety margin of only 0.20 from the fully honest state. As a rises, x^* climbs monotonically towards $x = 1$, so this margin closes continuously—from 0.20 towards zero—while published quality appears unchanged, because the population remains above the threshold throughout. By the time a reaches $a^* \approx 0.50$, the margin has vanished and the system tips into Global Corruption with no visible warning in quality metrics. While observable quality remains high, structural fragility grows rapidly—and this gap is precisely what makes AI-induced tipping difficult to detect in advance. We examine the testability of these predictions in Section 6.

Agent-based simulations confirm that the three main results—gradual onset of fragility, discrete tipping, and severe path dependence—survive finite-population noise and heterogeneous AI adoption diffusion (Appendices B and C of the Supplementary Material). Under a linear AI adoption ramp from $a = 0$ to $a = 0.8$, the ABM reproduces the ODE's tipping dynamics: the safety margin $\Delta = x_t - x^*(a_t)$ operates as a binary shield, with a buffer of even 0.02 preventing collapse across all community sizes tested ($N \in \{200, 500, 1{,}000\}$), while the absence of a margin exposes roughly half of all runs to quality decline regardless of N (Bonabeau, 2002; Traulsen and Hauert, 2009). Robustness analysis across the $(\delta/\gamma, \mu)$ plane reveals that the false-positive slope μ is the binding constraint: even if review capacity scales perfectly with submission volume ($\delta/\gamma = 1$), the system remains in Global Corruption whenever $\mu > 0.09$ (Checco et al., 2021). This confirms that the two policy levers—raising K_{rev} and reducing λ—must be deployed together, not sequentially.

5 Institutional Responses

We now turn from analysis to institutional design. The three results from Section 4 reduce the institutional problem to a single condition: keep $\Pi_{bad}(x, a)$ negative. AI adoption pushes Π_{bad} upward through four channels simultaneously: submission volume rises, review capacity lags, submission cost falls, and λ increases. The model suggests that effective policy must counteract each channel. We organise the response around five principles, ordered by the tightness of the constraint each relieves.

Principle 1: Scale verification ($K_{rev} \uparrow$). Higher review capacity raises the screening intensity $\eta = \min(1, K_{rev}/V)$, which reduces the bad-paper pass rate π_0 and lowers Π_{bad} at the corrupt boundary $x \approx 0$. Practical mechanisms include structured reviewer panels with explicit recognition credits (Aczél et al., 2021), AI-assisted consistency and format pre-screening that frees expert time for substantive judgement (Checco et al., 2021), and co-review programmes that train junior reviewers alongside senior ones. However, there is a binding caveat: when $\lambda > c(a)/r$, even perfect screening ($\eta = 1$) cannot make $\Pi_{bad}(1) < 0$ at the honest boundary. At $a = 0.8$ under baseline parameters, $\lambda = 0.26$ while $c(a)/r = 0.17$; no feasible increase in K_{rev} alone can stabilise the system at $K = 400$. Crucially, review assistance tools must not themselves raise λ: format and consistency checks are appropriate, but automated quality scoring is not (Naddaf, 2025).

Principle 2: Strengthen the screening signal ($\lambda \downarrow$) — the binding constraint. The GH condition at the honest boundary requires $\Pi_{bad}(1) < 0$, which holds if and only if $\lambda < c(a)/r$. This is the binding constraint: no amount of K_{rev} expansion or slot restriction restores GH while λ exceeds this threshold. Three mechanisms directly reduce λ. First, mandatory data, code, and materials sharing makes fabrication costlier and enables post-publication verification (Munafò et al., 2017; Nosek et al., 2015). Second, structured reporting checklists—CONSORT for randomised trials (Moher et al., 2009), PRISMA for systematic reviews—force authors to document methods in ways that make gaps visible to reviewers. Third, Registered Reports commit authors to their hypotheses and analysis plan before results are known, eliminating outcome-selective reporting (Nosek et al., 2018), which is itself a leading source of inflated false-positive rates (Simmons et al., 2011). Taken together, these mechanisms address the root cause that λ is elevated: reviewers cannot easily distinguish genuine contributions from polished but uninformative manuscripts.

Principle 3: Add smart friction ($c \uparrow$). The GH condition also requires $c > r \cdot \lambda$. As AI adoption lowers $c(a) = c_0(1 - \phi a)$, this gap closes and eventually inverts. Submission tokens or annual per-journal quotas raise the effective cost of marginal submissions without requiring any detection of AI use (Cotton, 2013). Escalating resubmission fees impose additional costs on the low-quality cycling of rejected papers through multiple journals. We note an equity caveat: blunt friction raises costs for all authors, including

those at under-resourced institutions; targeted exemptions are discussed in the final paragraph of this section.

Principle 4: Decouple dissemination from certification. The current publishing architecture conflates two distinct functions: disseminating findings quickly and certifying quality carefully. When certification degrades under submission pressure, both functions fail together. Separating the two—via preprint servers combined with overlay journals or community platforms such as Peer Community In (Ginsparg, 2011)—allows dissemination to remain immediate while certification proceeds at the pace required for genuine scrutiny. The slot-expansion paradox established in Section 4 applies specifically to certified publication slots K_{cert}, not to preprint availability. Decoupling therefore prevents the paradox from emerging: adding preprint capacity does not raise Π_{bad}, because preprint posting is not the payoff-generating event in the spam incentive.

Principle 5: Reshape evaluation rewards (r reduced for marginal output). The spam payoff scales linearly with r: $\Pi_{bad} = r \cdot \bar{\rho} \cdot \pi_0 - c$. Current career and grant structures treat paper counts as a primary productivity metric (Smaldino and McElreath, 2016; Casadevall and Fang, 2014), in effect maximising the reward for any accepted paper regardless of quality. Impact-weighted or reproducibility-indexed assessment would reduce the effective r for marginal submissions while preserving incentives for high-quality work (Stephan, 2012). The GH condition at $x = 1$ requires $\lambda(a) < c(a)/r$; reducing r directly widens this margin, enlarging the parameter region in which the honest long-run outcome is self-sustaining. Reward reform operates on a longer horizon than Principles 1–4 and requires coordination across funders, institutions, and evaluation bodies; it is therefore a deep-redesign lever rather than a quick win.

Figure 7 illustrates the joint effect of the principles in the (K_{rev}, K) parameter space (a detailed mapping of each mechanism to its primary parameter shift is in Appendix G of the Supplementary Material). The left panel of Figure 7 shows that at $a = 0.8$ under baseline parameters, Global Corruption dominates most of the space, and Set A sits deep in the GC region; the navy arrow shows that raising K_{rev} alone (Principle 1) leaves Set A in GC because $\lambda = 0.26 > c/r = 0.17$. The right panel applies Principles 2 and 3 jointly ($\lambda = 0.05$, $c = 35$, with $\phi = \mu = 0$ locking in those improvements): the GC region is eliminated entirely and Set A shifts to the bistability (BS) region; the dark green arrow then shows that combining Principle 1 (K_{rev} raised to 600) carries Set A into the Global Honesty region. The lesson is that Principles 2 and 3 are necessary to exit GC, while Principle 1 converts bistability into global honesty: the three levers are complements, not substitutes.

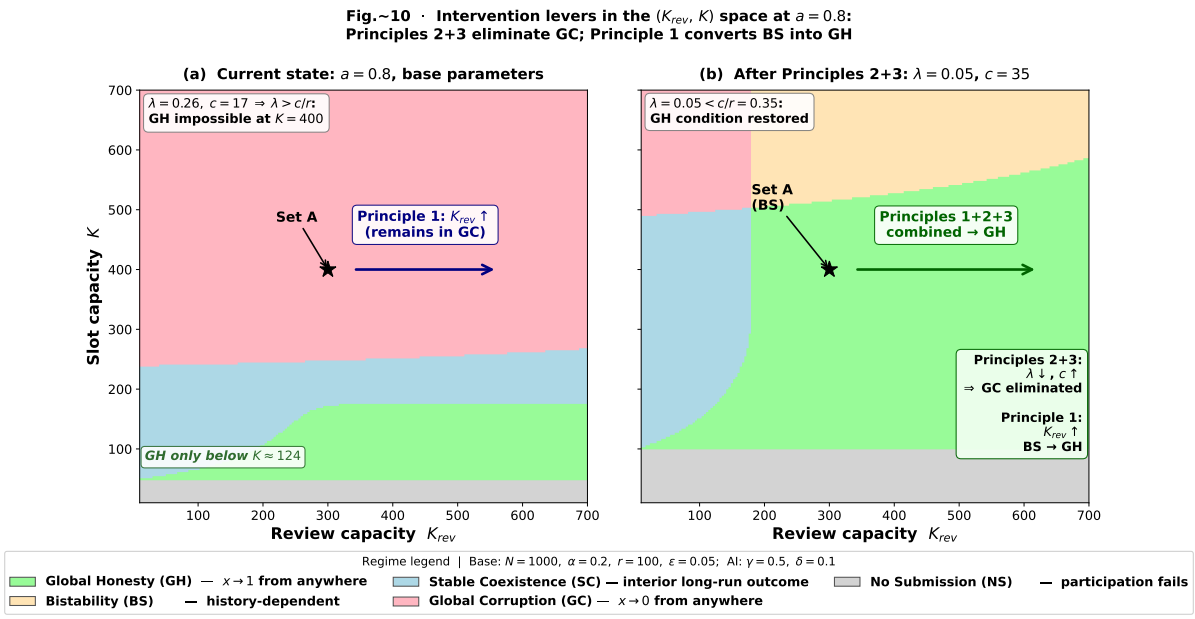

Figure 7: **Intervention levers in the (K_{rev}, K) parameter space at $a = 0.8$.** *Left panel:* base parameters ($\lambda = 0.26$, $c = 17$); GC dominates and Set A (star) sits deep in GC. The navy arrow shows Principle 1 ($K_{rev} \uparrow$) alone stays in GC because $\lambda > c/r$. *Right panel:* after Principles 2 and 3 ($\lambda = 0.05$, $c = 35$, $\phi = \mu = 0$); GC is eliminated and Set A shifts to BS. The dark green arrow shows the combined intervention (Principles 1+2+3, K_{rev} raised to 600) restores Global Honesty. Parameters: $\gamma = 0.5$, $\delta = 0.1$; Set A baseline $N = 1000$, $\alpha = 0.2$, $r = 100$, $\varepsilon = 0.05$.

Transition policy: resetting after tipping. Hysteresis means that gradual, journal-by-journal reforms are insufficient once the system has tipped into Global Corruption. In the GC long-run outcome, $\Pi_{bad} > 0$ for all values of the honest-author share x; no individual author has an incentive to deviate back towards honesty (Scheffer et al., 2001). Recovery requires a large, coordinated intervention that simultaneously restores λ, raises c, and expands K_{rev} enough to shift the entire parameter configuration out of the GC region. The ClinicalTrials.gov registration mandate, which required prospective trial registration as a condition of journal publication, illustrates the scale of coordination needed (Moher et al., 2009): a single policy change, applied across many journals simultaneously, moved an entire field. The implication for AI-era publishing is that funders and professional societies must act across journal families, not journal by journal.

Equity. Friction mechanisms ($c \uparrow$) and strict verification requirements ($\lambda \downarrow$) impose larger burdens on authors at under-resourced institutions, who may lack the data-sharing infrastructure, statistical support, or time to meet demanding new standards. Smart design can preserve the incentive effects while redistributing the cost: submission token grants for authors at lower-income institutions (Cotton, 2013); structured reporting checklists, which carry low infrastructure cost, preferred over mandatory open data repositories, which carry high infrastructure cost; subsidised reviewer development programmes that build review capacity where it is currently thin (Aczél et al., 2021). These equity adjustments do not change the core model predictions—the regime boundaries shift on the same dimensions—but they change who bears the adjustment cost.

Policy menu: quick wins and deep redesign. Some mechanisms require no inter-journal coordination and can be adopted immediately: structured reporting checklists, mandatory data and code sharing policies, submission token caps, and escalating resubmission fees. Others require collective action across journals or fields: overlay certification architectures, cross-journal quota systems, and coordinated field-wide mandates. The right sequencing strategy is to deploy the quick wins first, to arrest the upward drift of Π_{bad}, while building the coordination capacity needed for deep redesign before any tipping event occurs. Waiting for a quality crisis before acting is the worst strategy: recovering from GC requires a larger simultaneous push than preventing the tip in the first place (Scheffer, 2009).

The welfare analysis (Appendix D of the Supplementary Material) formalises the normative case for

these interventions. Spam submissions impose two unpriced externalities: a *congestion externality* that degrades review quality for all authors by tightening $\eta = K_{rev}/V$, and a *certification externality* that dilutes the precision of the certified pool, reducing its value for all downstream readers (Akerlof, 1970; Ostrom, 1990). Reducing λ is a Pareto improvement: it raises certification precision and honest-author surplus without imposing costs on authors who already meet the new standard (Corollary W4 in the Supplementary Material). Raising K_{rev} is welfare-improving when the congestion constraint binds, with costs borne by funders rather than authors (Corollary W2). Expanding K without reducing λ is the unique lever that unconditionally reduces welfare, because it raises Π_{bad} and degrades the certification signal with no offsetting benefit (Corollary W4). The normative lever ranking is therefore: $\lambda \downarrow$ first (Pareto-improving), then $K_{rev} \uparrow$ (public-good rationale), then $c \uparrow$ with equity exemptions (a corrective levy on the congestion externality), and never $K \uparrow$ in isolation. This welfare ordering provides the formal justification for the priority sequencing detailed in Appendix G of the Supplementary Material.

6 Empirical Predictions and Measurement

Prediction 1: Quality degradation is nonlinear—quiet, then a cliff. The bistable structure of the model generates a specific observable signature: quality metrics should hold approximately steady while AI adoption a rises below the tipping threshold, then deteriorate rapidly once the tipping threshold $x^*(a)$ reaches the current honest-author share. Smooth, continuous quality decline would be inconsistent with the model; the bistable structure produces a cliff, not a slope. The early-warning indicators are not quality itself but the shrinking gap between the current author composition x_t and the rising tipping threshold $x^*(a_t)$—a gap that narrows silently while quality appears stable (Scheffer, 2009). Empirical evidence is consistent with this reading. Monthly arXiv submissions rose approximately 50% between late 2022 and late 2024, while reviewer acceptance rates and post-publication retraction counts have not yet stabilised at new equilibrium levels (arXiv, 2024; Van Noorden, 2023). Field-level heterogeneity provides additional structure: Kobak et al. (2025) find AI-modified text rates exceeding 40% in some biomedical sub-disciplines while remaining near zero in others, suggesting that different fields may sit at different distances from their respective tipping thresholds.

Prediction 2: The gap between collapse and recovery thresholds widens as AI adoption rises. As a increases, Proposition 3 shows that the defence-by-congestion mechanism requires progressively tighter slot constraints to hold the corrupt boundary, while Proposition 1 shows that the Global Honesty boundary erodes. The model predicts that the intervention required to restore quality in a field that has tipped is strictly larger than the intervention that would have prevented the tip. Pre-AI evidence suggests that the hysteresis mechanism itself is empirically real, independently of what triggers the initial collapse. Bik et al. (2016) document that image duplication—driven by paper mills, not generative AI—became entrenched in specific biomedical sub-fields despite sustained editorial intervention; Van Noorden (2023) report that annual retraction counts exceeded 10,000 for the first time in 2023 without receding, suggesting that the low-quality long-run outcome sustains itself once established. The pre-AI paper-mill literature thus confirms the mechanism: quality collapses are not self-correcting, and removing the proximate trigger (the mills) does not restore the parameter configuration that originally supported honest submission. The AI-specific prediction is that the same mechanism will be triggered at lower adoption thresholds and across more fields simultaneously, because AI lowers $c(a)$ and raises $\lambda(a)$ for the entire author population rather than for a subset of bad actors. A direct test would compare the institutional effort required to restore quality indicators in a field that tipped—whether by paper mills before 2022 or by AI-driven flooding after—against one that never tipped but faces equivalent submission pressure.

Appendix E of the Supplementary Material develops four additional predictions: observable proxies for a rising false-positive rate (acceptance rates, surface polish, replication rates), proxies for review-capacity stress (reviewer acceptance rates, turnaround times, desk rejections, reviewer concentration), a data-source and identification strategy exploiting the November 2022 ChatGPT release as a natural experiment, and falsification conditions for the asymmetry hypothesis.

7 Discussion

Model boundaries and what they mean for interpretation. The model makes two structural choices that bound its applicability. First, we reduce the strategy space to two options: submit only good work, or submit regardless of quality. Real authors calibrate effort continuously, and intermediate strategies can coexist in finite communities. We justify the binary reduction by an evolutionary instability argument (Weibull, 1995): at any population state x, a strategy that adds bad-paper submissions earns strictly higher fitness when $\Pi_{bad} > 0$, and strictly lower fitness when $\Pi_{bad} < 0$, so intermediate submission rates are unstable under replicator dynamics and the population is driven to one of the two extremes in the mean-field limit. In practice, a continuum of effort levels would smooth the tipping threshold into a region rather than a sharp boundary, making early warning signals harder to detect but not eliminating them. Second, we model a single journal with fixed capacity. In reality, rejected papers cascade down a prestige hierarchy: Siler et al. (2015) find that roughly 62 % of papers rejected at three high-impact journals were eventually published elsewhere. This cascade distributes submission pressure across tiers rather than absorbing it. A multi-journal extension would predict that tipping first occurs at lower-tier venues—which typically have smaller K_{rev} and weaker screening—and then propagates upward as the pool of credibly certified papers shrinks. Our single-journal results are therefore conservative: the threshold at which quality degrades across the system as a whole is likely lower than the threshold we identify for any single venue.

The problem is institutional mismatch, not AI adoption itself. The model establishes that the core vulnerability is the gap $\gamma - \delta > 0$, not AI adoption per se. If review capacity scaled at the same rate as submission volume—$\delta = \gamma$—and AI tools improved screening reliability rather than degrading it—$\mu \leq 0$—the Global Honesty region would expand under rising a, not contract. Noy and Zhang (2023) document genuine productivity gains from AI assistance; the issue is not that AI exists but that the gap between production and verification growth rates exists. Acemoglu and Restrepo (2019) provide a useful framing: automation raises productivity in tasks that are routine and codifiable while increasing demand for tasks requiring contextual judgement and domain expertise. While text generation is largely the former, verifying the validity of a novel empirical claim is largely the latter. The challenge for publishing institutions is therefore not to restrict AI adoption—which would be both unenforceable and wasteful where AI genuinely helps—but to close the gap between γ and δ through targeted investment in verification infrastructure. Which scenario a field occupies is an institutional choice, not a technological fate. Fields that invest in structured AI-assisted verification pipelines, mandatory data deposition, and reviewer development programmes can remain stable even as a rises. Fields that do not will drift towards collapse (see the scenario matrix in Appendix G of the Supplementary Material).

We develop several model extensions—multi-journal competition and quality cascades, heterogeneous AI adoption, and endogenous reviewer effort—in Appendix F of the Supplementary Material.

8 Conclusion

We have studied what happens to a publishing system when AI raises the productivity of manuscript production faster than it raises the capacity to verify manuscripts. The central asymmetry—production sensitivity γ exceeding verification sensitivity δ—is not a modelling assumption but an empirical claim supported by documented submission growth, documented reviewer decline, and experimental evidence on AI writing productivity (Noy and Zhang, 2023; arXiv, 2024; Meyerson et al., 2025). Three results follow from this asymmetry. First, extreme slot scarcity can stabilise the honesty norm under degraded screening, but AI erodes this protection continuously by lowering the submission cost $c(a)$: the required slot tightness grows with adoption and eventually forces the system into the No Submission zone. Second, gradual AI adoption can trigger a discrete collapse rather than a smooth decline; the system can appear healthy—with observable quality near its maximum—while the gap between the current author composition and the tipping threshold narrows without any visible warning in quality metrics. Third, once the system tips, recovery is harder than prevention, because the low-quality equilibrium is self-sustaining: no individual author has a reason to deviate back towards honesty, and small interventions cannot overcome the strategic lock-in (Scheffer, 2009). Expanding publication slots accelerates rather

than relieves this dynamic, because it raises the expected payoff of submitting low-quality work without addressing the underlying screening deficit.

Translating these results into institutional design requires targeting the parameters that matter most. The probability λ that weak papers pass review is the binding constraint: when $\lambda > c(a)/r$, no feasible increase in review staffing can restore the Global Honesty condition, because even carefully reviewed weak manuscripts pass at a rate that makes submission individually rational. The model suggests that effective policy must reduce λ directly—through mandatory data and code sharing, structured reporting requirements, and registered study designs—while simultaneously raising K_{rev} and restoring submission friction (Munafò et al., 2017; Cotton, 2013). These three levers are complements, not substitutes: our analysis in Section 5 shows that λ reduction is necessary to exit the Global Corruption region, and K_{rev} expansion is necessary to convert bistability into Global Honesty. The institutional response to AI is therefore not "more slots" but a different kind of institution—one in which verification is a first-class resource, not an afterthought.

The window for preventive action is narrower than it appears. Because the system can remain near the honest equilibrium while structural fragility accumulates, the observable quality signal gives no reliable warning before a tipping event (Scheffer et al., 2001). While fields that invest in verification infrastructure, reviewer incentives, and submission friction before AI adoption deepens can preserve the integrity of certification at moderate cost, fields that wait for a visible quality crisis will face a recovery problem that requires a larger and more disruptive coordinated intervention—the equivalent of a field-wide mandate rather than a journal-level policy. The asymmetry between production and verification is not temporary: it reflects a fundamental difference in what language models can automate, and institutions designed for an earlier era will need deliberate redesign to remain effective.

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